

Draft - expert review pending

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Compiled with problems:

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

ERROR in ./docs/semester-1/ggur-300/unit-03/topic-01.mdx

Module build failed (from ./node_modules/@docusaurus/mdx-loader/lib/index.js):

Error: MDX compilation failed for file

"/home/a2ahs/mega book for B.Ed/.claude/worktrees/agent-affcb48d7818e8183/docs/semester-1/ggur-300/unit-03/topic-01.mdx"

Cause: Unexpected character `=` (U+003D) before name, expected a character that can start a name, such as a letter, `\$`, or `\_`

Details:

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"file": "",

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"line": 141,

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},

"reason": "Unexpected character `=` (U+003D) before name, expected a character that can start a name, such as a

letter, `\$`, or `\_`"

2. Which school task needs quantitative reasoning rather than bare

"ruleId": "unexpected-character",

"source": "micromark-extension-mdx-jsx",

"url": "https://github.com/micromark/micromark-extension-mdx-jsx#unexpected-character-at-expected-expect"

}

3. A headteacher says: "63 pupils signed up for games period and the ground

fits 80." A quantitatively reasoning teacher should first: (Apply) a) accept

at compileToJSX

the plan because 63 is less than 80 b) check the arithmetic and ask what other quantities the decision ignores c) refuse the plan until a calculator is available d) ask the pupils to count themselves

4. Steen's argument for quantitative literacy rests mainly on: (*Understand*) a) employers demanding faster clerks b) democratic life arriving as numbers that citizens must be able to question c) universities needing more mathematicians d) mathematics examinations becoming harder
5. Number sense is best described as: (*Remember*) a) memorising number facts up to twenty b) a feel for the size, meaning, and relationships of numbers c) writing numbers neatly and correctly d) counting objects quickly
6. A trainee estimates 47×52 as $50 \times 50 = 2,500$. The strategy used is: (*Apply*) a) front-end digits b) rounding to compatible numbers c) grouping d) order-of-magnitude
7. The school must file its official enrolment return to the district office. The appropriate approach is: (*Apply*) a) an estimate based on last year's figure b) an exact count from the attendance register c) an approximation to the nearest ten d) a count of one class multiplied by the number of classes
8. The four steps of the problem-solving strategy, in order, are: (*Remember*) a) plan, understand, check, compute b) understand, plan, carry out, look back c) compute, check, plan, understand d) understand, compute, plan, look back
9. "If the van seats 30, and 34 pupils signed up, then one trip cannot carry everyone." This sentence is an example of: (*Apply*) a) elimination b) if-then reasoning c) pattern spotting d) approximation
10. In the bench problem, the look-back step changed the answer from 60 benches to 62 because: (*Apply*) a) the division 240 by 4 had been done wrongly b) real-world needs (chief guest and teachers) were missing from the bare computation c) benches sometimes break and spares are required by rule d) 60 is an unlucky number for functions

Restricted-response questions (RRQs)

1. Define quantitative reasoning in one sentence of your own, and give one example from school life. (*Understand*)
2. Explain in two sentences why a teacher's attitude towards numbers matters for pupils. (*Understand*)
3. A colleague says: "Quantitative reasoning means doing hard calculations." Correct the claim in two sentences, using the idea that reasoning serves a decision. (*Analyze*)
4. State the difference between an estimate and an approximation, with one example of each. (*Understand*)
5. A school has 348 pupils and each pupil needs 2 worksheets; worksheets come 10 to a packet. Estimate the number of packets needed, and name the strategy you used. (*Apply*)
6. For each situation, say whether an exact figure or an estimate is appropriate, and why: (a) ordering exam papers for Class 8; (b) deciding whether the school van can carry all pupils in one trip. (*Analyze*)
7. Explain in two sentences why "about 800 people" was good enough for the water contractor at sports day, but the official enrolment return must be an exact count. (*Understand*)
8. Name the four logical-reasoning moves, and give a one-line example of any two of them. (*Understand*)
9. A water tank holds 500 litres and the school uses about 100 litres a day. State the if-then conclusion, and name one condition that would break it. (*Apply*)
10. "Attendance was 90 percent on Monday and 90 percent on Tuesday, so the same pupils were absent on both days." Identify the flaw in this reasoning in two sentences. (*Analyze*)

Extended-response questions (ERQs)

1. Your headteacher announces: "Next month we will combine Classes 4 and 5 for games period because 63 pupils have signed up and the ground fits 80." Write a short note to the headteacher that checks the arithmetic claim, names one further quantity the decision ignores, and recommends whether to go ahead, with reasons. (*Analyze*)
2. The Parent Teacher Association can spend 25,000 rupees on prizes for a school of 512 pupils, and the president plans 100 prizes at about 250 rupees

each. Estimate whether the plan fits the budget, identify one quantity the plan ignores that could change the conclusion, and recommend the plan or a change to it, with reasons. (*Analyze*)

3. A school notice says: "Last year 20 pupils failed the Class 8 math exam. This year we bought a new workbook series, and only 10 pupils failed. Therefore the new workbooks halved our failure rate." Analyse the claim: name the reasoning move used, identify at least two other quantities that changed between the years and could explain the drop, and state what evidence would be needed before crediting the workbooks. (*Analyze*)
4. At sports day, Miss Farzana estimated the crowd by counting one section (about 40 people) and multiplying by the 20 sections she could see. Reconstruct her estimate as the four-step problem-solving strategy, name the estimation strategy she used, and explain in what situation her method would not be good enough. (*Analyze*)
5. Using a real decision you observed at your practicum school that involved a number (from your Topic 1.1 practicum task or another), describe the decision, then analyse it with this unit's tools: what was asked and given, which reasoning moves were used, whether an estimate or an exact figure was appropriate, and whether the look-back step happened. Evaluate the quality of the decision and name one thing you would do differently. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. **b** - quantitative reasoning is reasoning with numbers in context for a decision; arithmetic speed is only a component.
2. **c** - printing decisions must weigh absentee patterns; the other tasks are pure procedure.
3. **b** - checking the claim and asking what else matters is exactly the two-part habit the unit teaches.
4. **b** - Steen's case is about citizenship: numbers arrive in public life and must be questionable.
5. **b** - number sense is the feel for size and meaning, not memorised facts.

6. **b** - 47 and 52 were rounded to the compatible numbers 50 and 50.
7. **b** - an official return is a legal record; only the register count is appropriate.
8. **b** - understand, plan, carry out, look back.
9. **b** - two stated facts force a conclusion; that is if-then reasoning.
10. **b** - the look-back step adds real-world requirements the bare computation omitted.

RRQ model answers and mark schemes

1. Model: quantitative reasoning is using numbers, data, and logic to understand a situation and make a sound decision; example: grouping 240 pupils into groups of 15 to plan a reading programme. **Marks (2):** 1 - definition in the learner's own words naming reasoning and purpose; 1 - a genuine school example.
2. Model: pupils copy their teacher's confidence or fear with numbers; a teacher who reasons aloud shows pupils that numbers serve decisions. **Marks (2):** 1 - the modelling idea; 1 - a concrete consequence for pupils.
3. Model: the correction should move the centre of gravity from calculation to judgement: the hard part is deciding what to calculate and what the result means for a decision; calculation is only a tool inside that judgement. **Marks (2):** 1 - names the decision focus; 1 - gives a contrasting example or a clear restatement.
4. Model: an estimate is a reasoned guess at a value (crowd at the mela, about 800); an approximation is a deliberately simplified value used in place of an exact one to make work easier (taking 398 as 400). **Marks (2):** 1 - the distinction; 1 - one example of each.
5. Model: $348 \times 2 = 696$ sheets; $696 / 10 = 69.6$, so about 70 packets (buy 70; some add a spare packet). Strategy: rounding (348 to 350) or direct computation with a named strategy; any defensible named strategy earns the mark. **Marks (2):** 1 - a correct estimate near 70 packets; 1 - a named strategy.
6. Model: (a) exact - each candidate needs a paper, and shortage blocks the exam; (b) an estimate can suffice for a one-trip decision if there is a clear margin, but an exact count is safer when the number is near the van's capacity. Full marks require linking the choice to the consequence of being

wrong. **Marks (4):** 1 - (a) exact; 1 - reason tied to consequence; 1 - (b) either defensible choice; 1 - reason tied to margin or consequence.

7. Model: the water decision only needed the right order of size (a few hundred), so an estimate served the purpose; the enrolment return is an official record where an error has consequences regardless of size. **Marks (2):** 1 - purpose-relative accuracy; 1 - the official-record point.
8. Model: pattern, if-then, deduction, elimination (order not required); examples such as "if the tank holds 500 litres and we use 100 a day, it lasts 5 days" (if-then) or "the book is not in the cupboard, not on the shelf, not borrowed; it is lost" (elimination). **Marks (3):** 1 - all four moves named; 2 - two correct one-line examples (1 each).
9. Model: if the tank is full and use is 100 litres a day, the water lasts about 5 days; the conclusion breaks if the tank is not full, if daily use is higher, or if the tank leaks. **Marks (2):** 1 - the if-then conclusion; 1 - one breaking condition.
10. Model: the two percentages describe different days and do not identify anyone; the same 90 percent could be produced by entirely different absentees, so the conclusion about "the same pupils" is not forced. **Marks (2):** 1 - identifies the unsupported identity claim; 1 - explains why equal percentages do not imply the same pupils.

ERQ rubrics

1. Games-period note. Analytic rubric: *Arithmetic checked* (0-2: verifies $63 < 80$ and probes what the claim assumes); *Missing quantity identified* (0-2, **Analyze**: names a real ignored quantity such as supervision ratio or equipment and explains its effect); *Recommendation justified* (0-2: the recommendation follows from both checks); *Clarity and register* (0-2: a respectful, plain note a headteacher would act on). Total 8.
2. PTA prize budget. Analytic rubric: *Estimate with named strategy* (0-2: $100 \times 250 = 25,000$ fits exactly, shown with a stated strategy); *Ignored quantity* (0-2, **Analyze**: names a real factor, e.g. prize tiers, transport, or that 100 prizes for 512 pupils means most pupils get nothing, and shows its effect); *Recommendation* (0-2: follows from the analysis); *Clarity* (0-2). Total 8.
3. Workbook claim. Analytic rubric: *Move named* (0-2: identifies the if-then cause-effect structure); *Alternative explanations* (0-3, **Analyze**: two or more real changed quantities, e.g. a different cohort, a different exam, extra

coaching, each with a sentence of mechanism); *Evidence standard* (0-2: names a comparison that would isolate the workbook effect, e.g. a comparable school without the workbooks); *Clarity* (0-1). Total 8.

4. Sports-day reconstruction. Analytic rubric: *Four-step reconstruction* (0-3: understand, plan, carry out, look back each correctly mapped to her actions); *Strategy named* (0-2: grouping / section-and-multiply); *Limit of the method* (0-3, **Analyze**: names a situation where it fails, e.g. unevenly filled sections, moving crowds, or a safety decision needing an exact figure, with a reason). Total 8.
5. Practicum decision analysis. Analytic rubric: *Decision described* (0-2: a real, specific school decision with its number); *Tool application* (0-3: asked/given, reasoning move, estimate-vs-exact all applied to the case); *Look-back evaluation* (0-2, **Analyze**: judges whether the decision survived its look-back and why); *Improvement named* (0-2: one concrete thing the learner would do differently); *Clarity* (0-1). Total 10.

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

Give feedback